Course Notes

Lecture 03 — Low-Power Wide-Area Network (LPWAN) Access Technologies

Communication Technologies for the Internet of Things

Academic year 2026–2027

Catarina Silva

ESTGA, University of Aveiro

Text to support independent study

Contents

How to Use These Notes **2**

1 Commentary on the Slides **2**

2. Agenda . . . 2
3. Why Long-Range, Low-Power Networks (1/2) . . . 3
4. Why Long-Range, Low-Power Networks (2/2) . . . 4
5. Positioning LPWAN in the Solution Space . . . 5
6. LoRa and LoRaWAN: Background . . . 6
7. LoRaWAN: Technical Characteristics (1/2) . . . 7
8. LoRaWAN: Technical Characteristics (2/2) . . . 8
9. LoRaWAN: Architecture and Device Classes (1/2) . . . 9
10. LoRaWAN: Architecture and Device Classes (2/2) . . . 10
11. LoRaWAN: Network Architecture . . . 11
12. LoRaWAN: Device Classes . . . 12
13. Sigfox: Overview (1/2) . . . 13
14. Sigfox: Overview (2/2) . . . 14
15. Sigfox in Portugal . . . 15
16. Weightless: Overview . . . 16
17. Weightless: Comparison of Variants . . . 17
18. RPMA (Ingenu) (1/2) . . . 18
19. RPMA (Ingenu) (2/2) . . . 19
20. EnOcean: Battery-Free Communication (1/2) . . . 20
21. EnOcean: Battery-Free Communication (2/2) . . . 21
22. Z-Wave (1/2) . . . 22
23. Z-Wave (2/2) . . . 23
24. Licensed and Unlicensed Spectrum . . . 24
25. LTE-M: Overview (1/2) . . . 25
26. LTE-M: Overview (2/2) . . . 26
27. NB-IoT: Overview (1/2) . . . 27
28. NB-IoT: Overview (2/2) . . . 28
29. NB-IoT: Modes of Operation (1/2) . . . 29
30. NB-IoT: Modes of Operation (2/2) . . . 30
31. 5G in Support of the Internet of Things (1/2) . . . 31
32. 5G in Support of the Internet of Things (2/2) . . . 32
33. Cost of Communication Modules (1/2) . . . 33
34. Cost of Communication Modules (2/2) . . . 34
35. Service Pricing Models (1/2) . . . 35
36. Service Pricing Models (2/2) . . . 36
37. Evolution Prospects (1/2) . . . 37
38. Evolution Prospects (2/2) . . . 38
39. Criteria for Selecting an LPWAN Technology . . . 39
40. Lesson Summary (1/2) . . . 40
41. Lesson Summary (2/2) . . . 41
42. Sources . . . 42

Final Check **43**

References **43**

How to Use These Notes

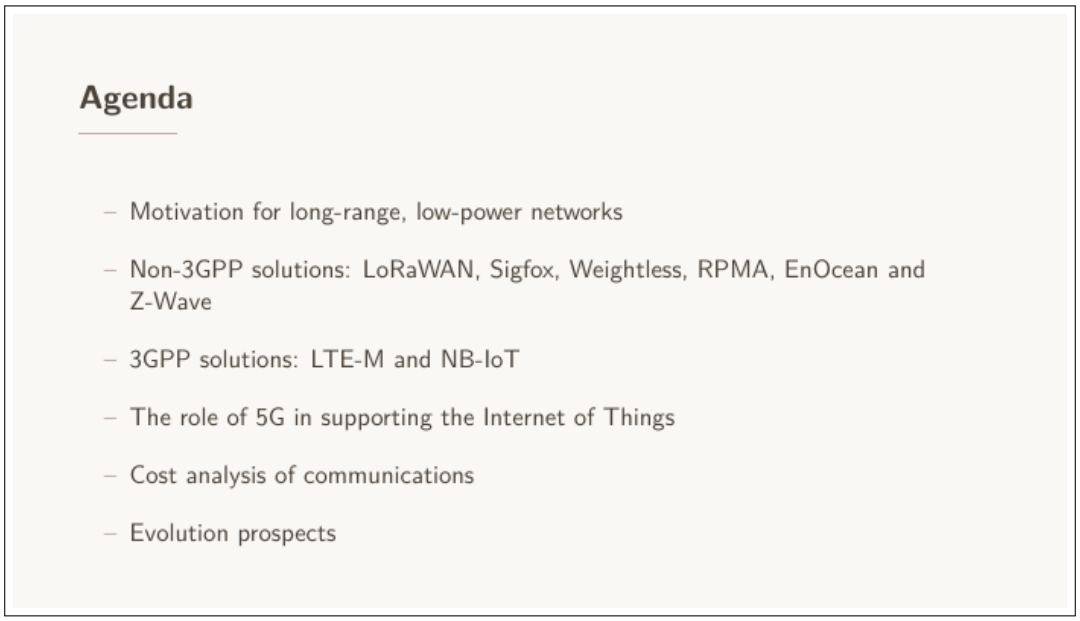
These notes follow the lecture slide by slide. Each page reproduces a slide and states, below it, what should be taken from it.

Reading them does not replace the lecture or the slides. They are meant for later study, once the student has heard the lecture and wants to consolidate, check that the essentials were understood or prepare for assessment. The comments make explicit links between slides that remain implicit in the oral presentation, and point out where the material is commonly misunderstood.

Whenever a comment poses a question, that question is the verification criterion: if you cannot answer it, go back to the slide before moving on.

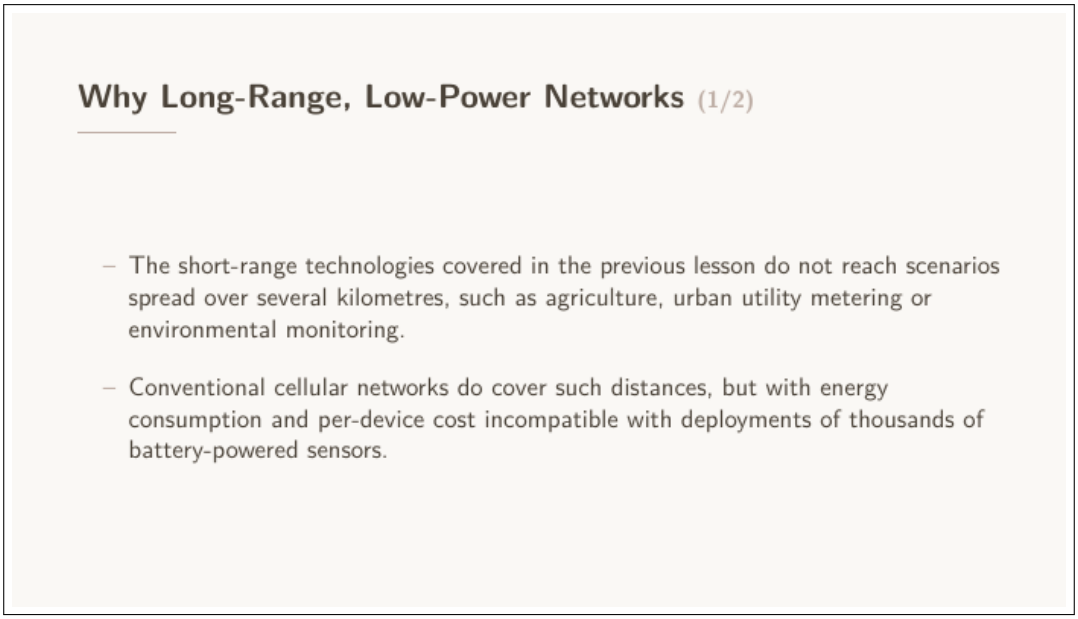
1. Commentary on the Slides

Slide 2. Agenda



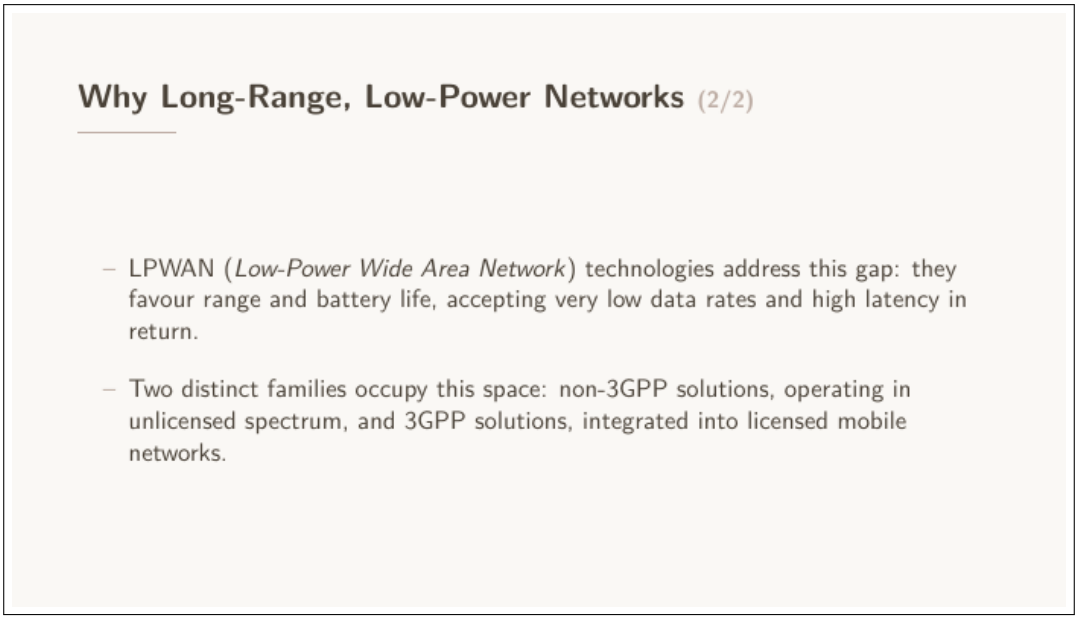
The path through the lecture. The sequence of topics is not arbitrary. It starts with the justification of the problem, goes through the solutions operating in free spectrum, moves on to those using licensed spectrum and ends with the economics of the decision. This order is worth retaining, because it matches the order in which questions arise in a real project: first you realise that no known technology fits, then you look for candidates, finally you compare the total cost of operating them.

Slide 3. Why Long-Range, Low-Power Networks (1/2)



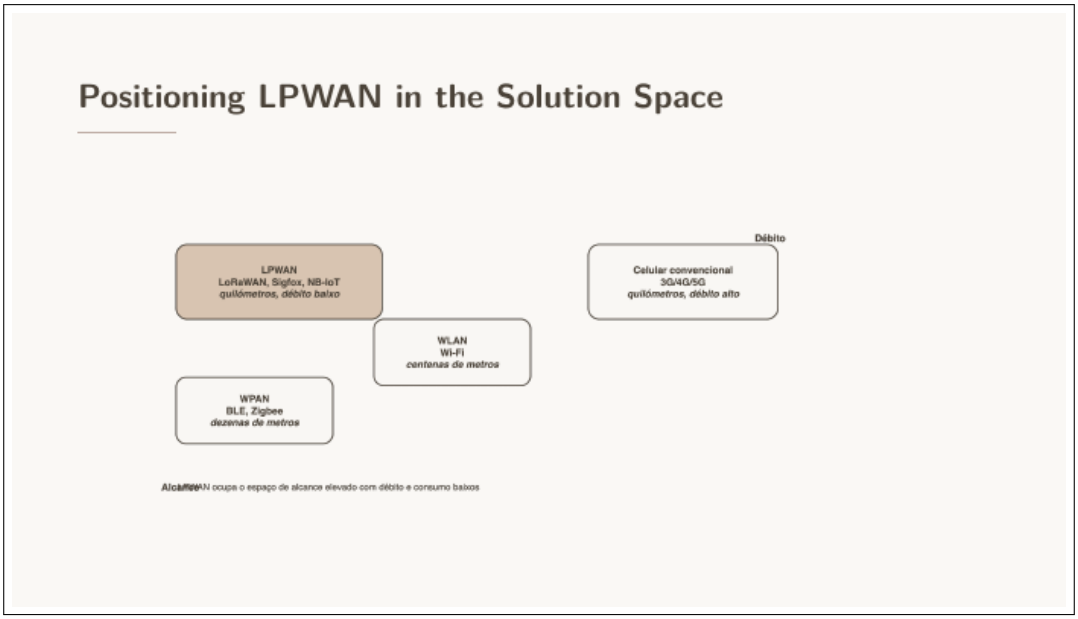
The problem that gives rise to LPWAN. This slide establishes the gap. The short-range technologies studied in the previous lecture cover tens of metres, and conventional cellular networks cover kilometres but require expensive, energy-hungry terminals. Between the two there is a void. The student should be able to name three concrete scenarios that fall in that void, and to explain, for each, which of the two families fails and why. Agriculture, meter reading and environmental monitoring are the examples used, but the reasoning generalises.

Slide 4. Why Long-Range, Low-Power Networks (2/2)



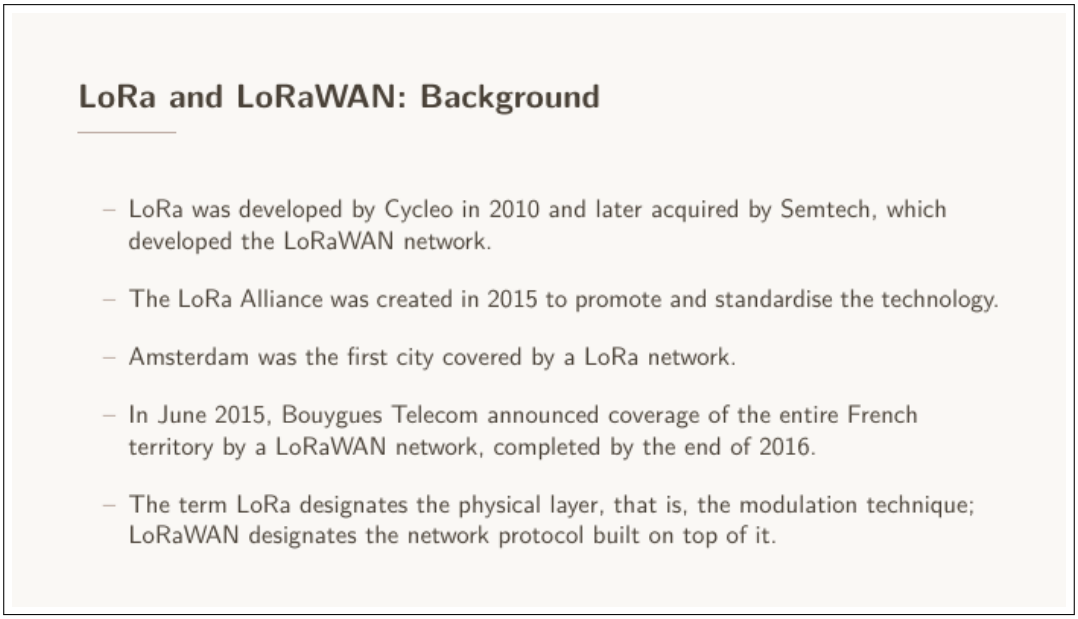
The answer: trading data rate for range and autonomy. The LPWAN definition presented here is deliberately given in terms of a compromise, not of a technology. An LPWAN favours range and battery life, and accepts in exchange very low data rates and high latencies. Retain also the division into two families, which structures the whole lecture: non-3GPP solutions, which operate in unlicensed spectrum, and 3GPP solutions, integrated into mobile networks. This distinction is not merely technical. It determines who operates the network, who answers for service quality and how traffic is charged.

Slide 5. Positioning LPWAN in the Solution Space



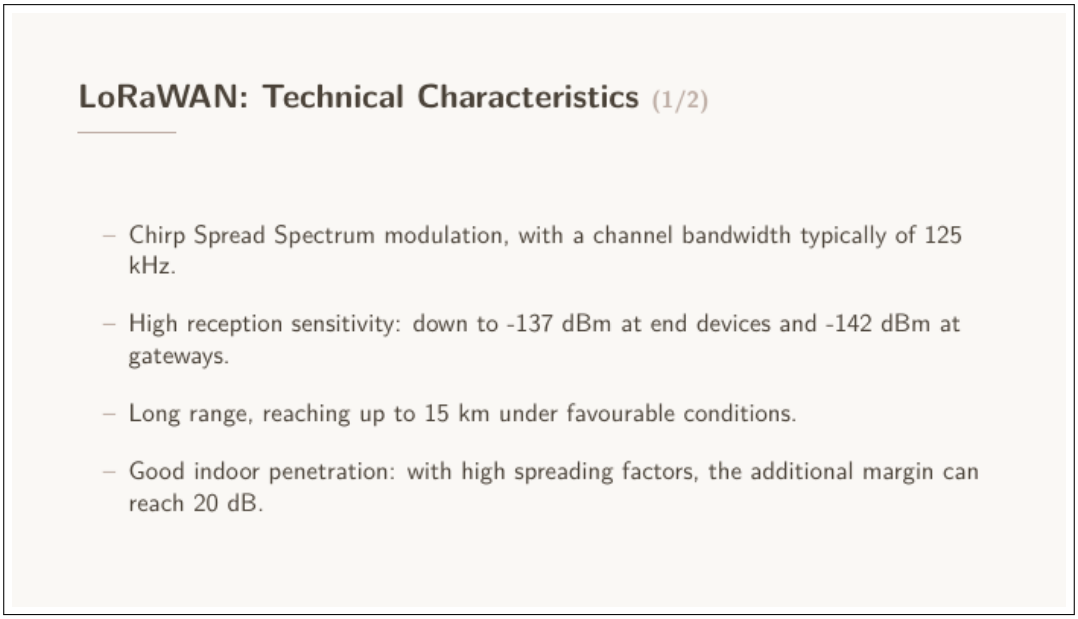
Placing LPWAN in the solution space. The diagram positions the four technology families on two axes, range and data rate. The useful reading is by exclusion: LPWANs occupy the region of high range with low data rate, precisely the region that none of the other families covers. When studying, it is worth placing on this diagram the technologies already known from earlier lectures, BLE and Zigbee, and checking that they fall in the opposite corner.

Slide 6. LoRa and LoRaWAN: Background



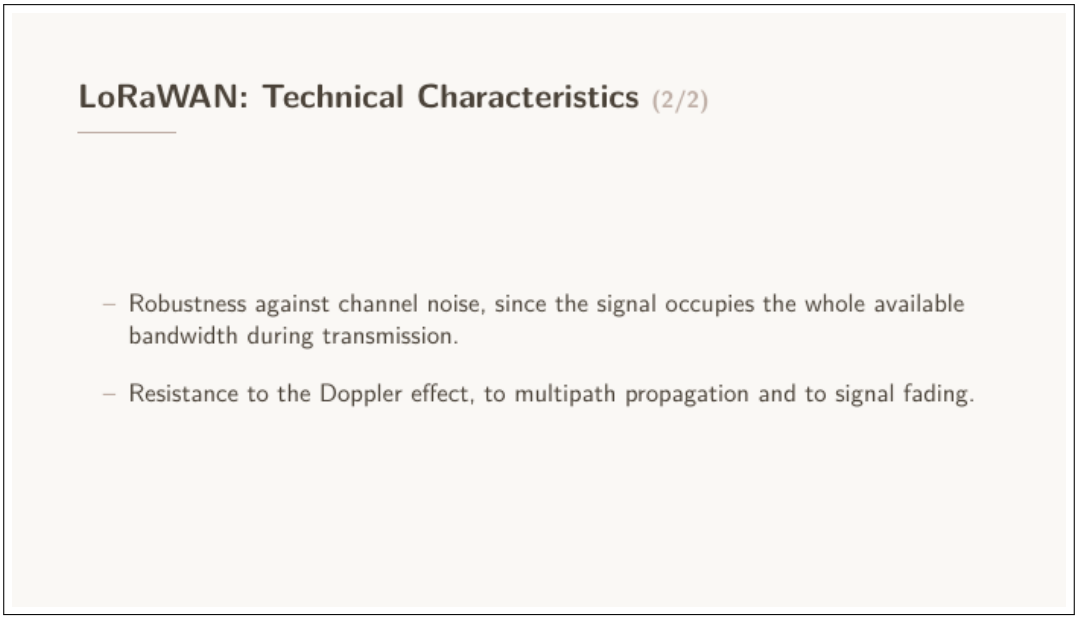
LoRa and LoRaWAN are not the same thing. The distinction between the two terms is often confused and is exam material. LoRa designates the physical layer, that is, the modulation technique, owned by Semtech. LoRaWAN designates the network protocol built on that physical layer, maintained by the LoRa Alliance. The practical consequence is relevant: it is possible to use LoRa radios without using LoRaWAN, building a point-to-point protocol of your own. The converse does not hold. The deployment history, with Amsterdam and the French national coverage, serves to show that the technology left the laboratory very early.

Slide 7. LoRaWAN: Technical Characteristics (1/2)



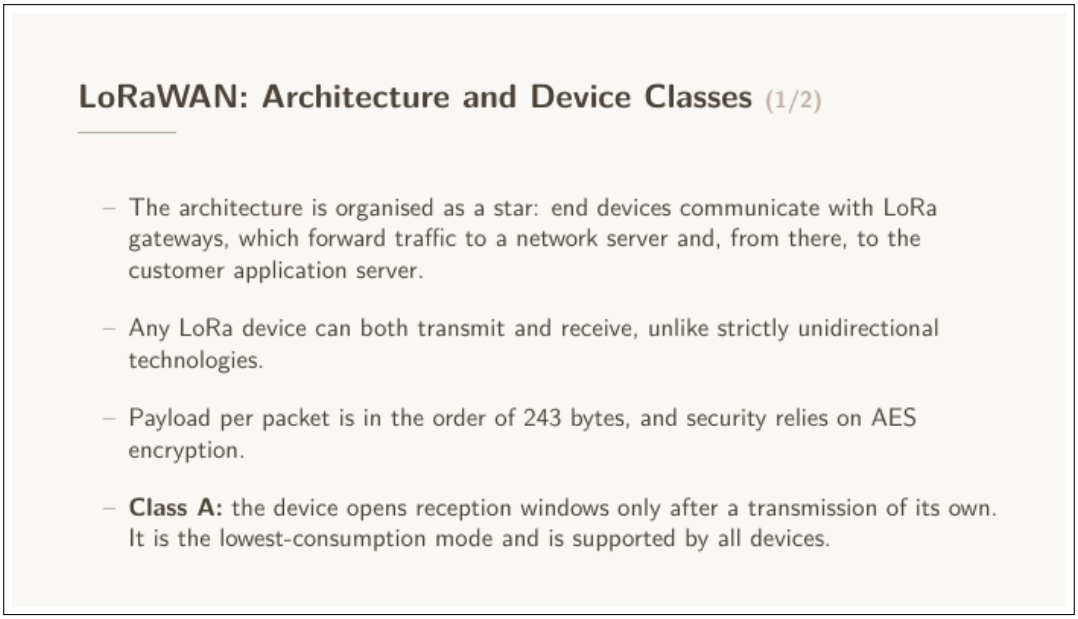
How modulation sustains range. Here the origin of the high range is explained. Chirp spread spectrum modulation spreads the signal across the whole channel bandwidth, which makes it detectable below the noise level. The sensitivities given, down to -137 dBm on terminals and -142 dBm on gateways, are the numerical expression of that property. The student should understand the cause and effect: it is not transmit power that gives range, it is the ability to receive in weak-signal conditions.

Slide 8. LoRaWAN: Technical Characteristics (2/2)



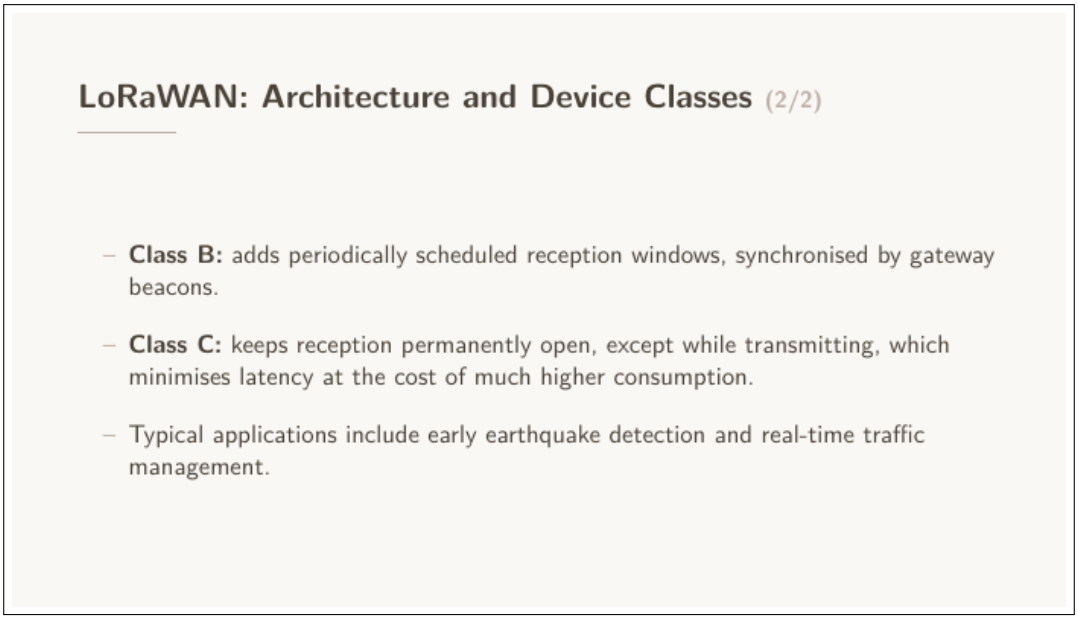
Channel robustness. Two effects deserve attention. The first is resistance to noise, a direct consequence of the signal occupying the whole bandwidth. The second is resistance to the Doppler effect and to multipath propagation, which makes the technology usable with moving nodes and in urban environments with many reflections.

Slide 9. LoRaWAN: Architecture and Device Classes (1/2)



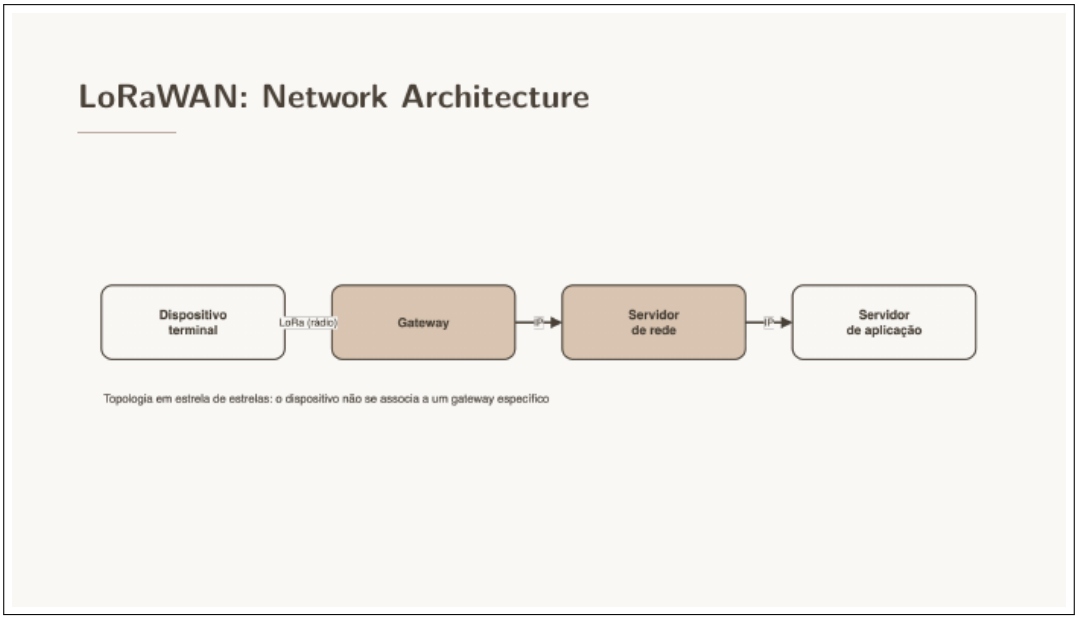
Star architecture and Class A. The LoRaWAN architecture is organised as a star, with devices communicating with gateways that forward to a network server. Note the difference from Zigbee: there is no routing between end devices, and the device does not associate with a particular gateway. Class A is the lowest-consumption mode and the only mandatory one. The device opens receive windows only after transmitting, which means the server cannot contact the device when it wants, only when the device gives it the opportunity.

Slide 10. LoRaWAN: Architecture and Device Classes (2/2)



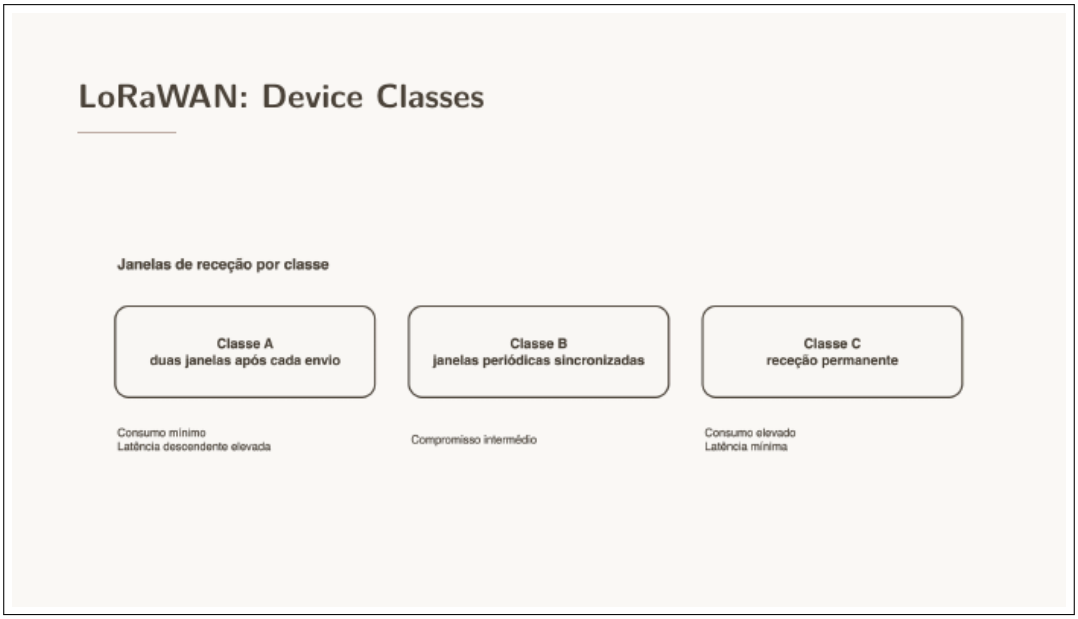
Classes B and C, and the price of availability. The three classes form a scale of compromise between consumption and downlink latency. Class B adds periodic windows synchronised by beacons. Class C keeps reception permanently open. The question to be able to answer is: given a downlink latency requirement, what is the minimum class that satisfies it, and what is the energy cost of that choice.

Slide 11. LoRaWAN: Network Architecture



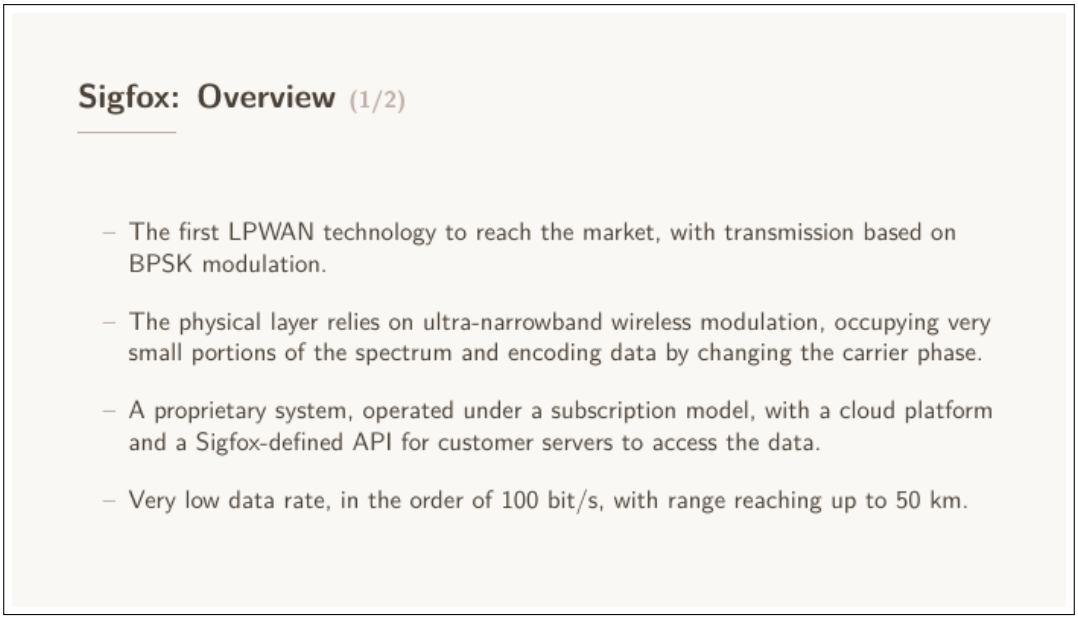
The complete chain, from device to application. The diagram shows the four elements of the architecture and the protocol used on each leg. The observation to retain is in the caption: the device does not associate with a specific gateway. The same transmission may be received by several gateways, and it is the network server that eliminates duplicates. This property has two consequences. It simplifies the device, which does not need to manage association or run handover procedures between cells. And it makes the network more robust, because the loss of a gateway does not isolate the devices that were within its range.

Slide 12. LoRaWAN: Device Classes



Visual comparison of the three classes. The diagram summarises the compromise. When studying, it helps to associate a plausible use case with each class: a water meter in Class A, a sensor with occasional commands in Class B, an actuator that must respond immediately in Class C.

Slide 13. Sigfox: Overview (1/2)



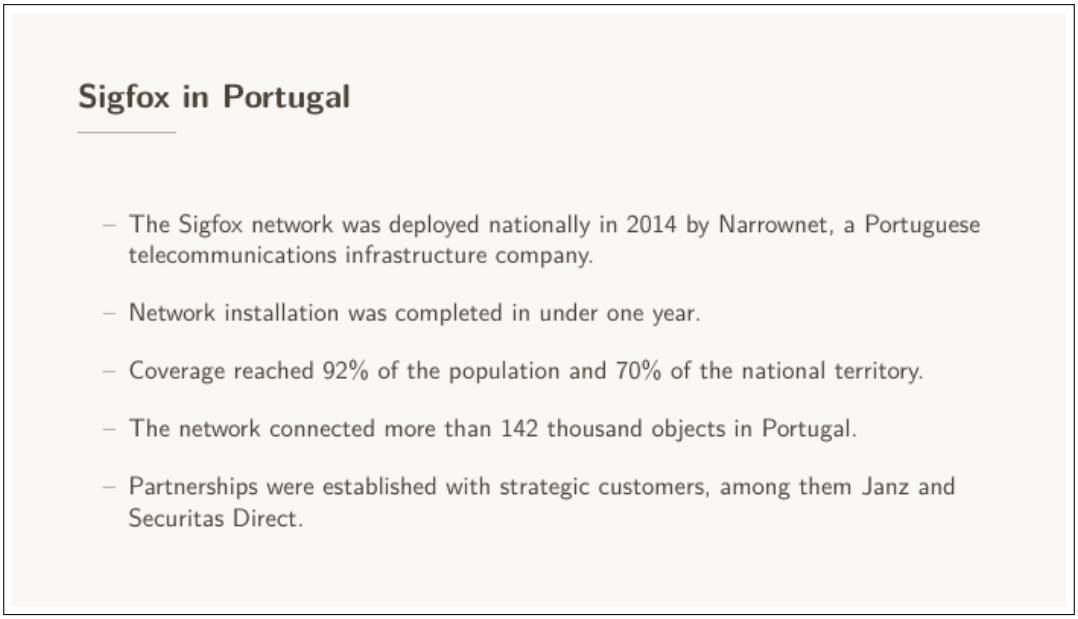
Sigfox: ultra narrow band and operator model. Sigfox adopts the opposite strategy to LoRaWAN regarding spectrum occupation. It uses ultra narrow band, occupying minimal portions, and reaches up to 50 km with a data rate in the order of 100 bit/s. Equally important is the operating model: a proprietary system, operated by subscription, with a cloud platform and data access by API. The customer does not operate the network. You buy connectivity, not infrastructure.

Slide 14. Sigfox: Overview (2/2)



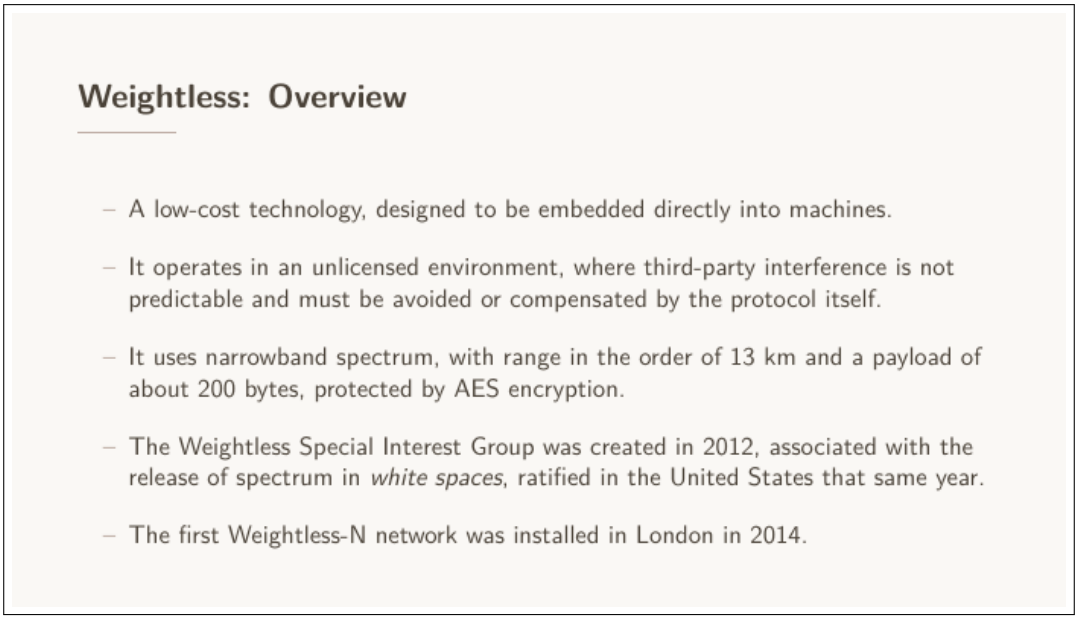
The constraints that define the use case. The limit of 140 messages per day and the payload of about 12 bytes are the numbers that determine whether Sigfox fits or not. One message every ten minutes already exceeds the limit. Note also the absence of security mechanisms of its own at the technology level, which transfers that responsibility to the application.

Slide 15. Sigfox in Portugal



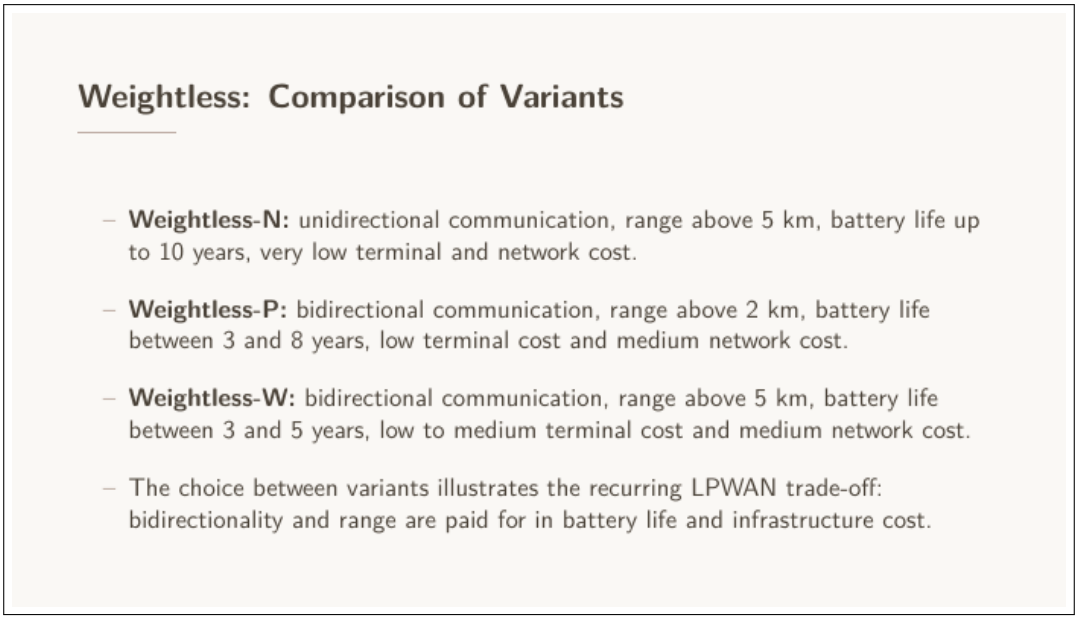
The Portuguese case. The national deployment by Narrownet in 2014, completed in less than a year and covering 92 per cent of the population, shows the speed of deployment that ultra narrow band allows. The example also serves to show that the availability of a technology is not uniform between countries, which is a decision criterion in itself.

Slide 16. Weightless: Overview



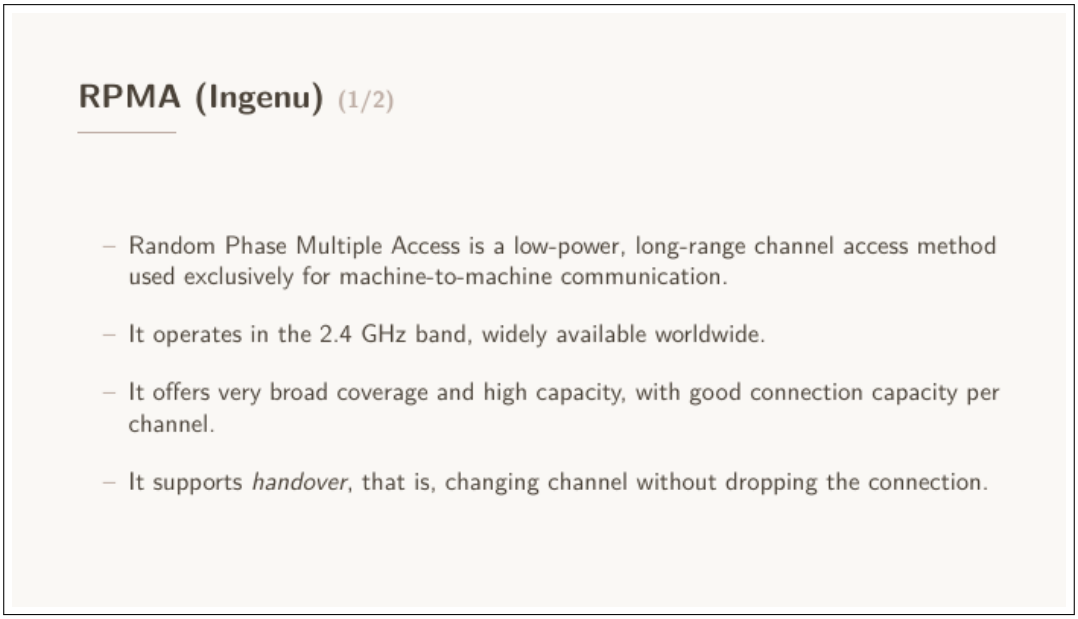
Weightless and the white spaces of the spectrum. The Weightless technology stands out for operating in white spaces, television spectrum bands not used locally. Operation in an unlicensed environment implies that interference from third parties is not predictable and has to be handled by the protocol itself.

Slide 17. Weightless: Comparison of Variants



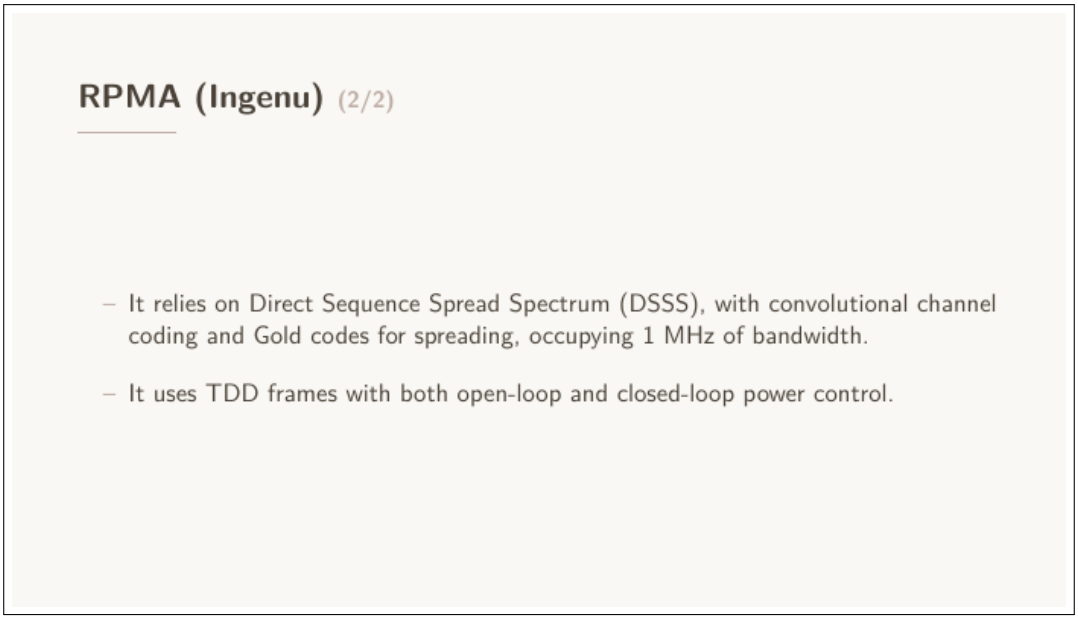
The three variants and the recurring trade-off. The N, P and W variants differ in directionality, range, autonomy and cost. The conclusion to draw is not the list of numbers, it is the pattern: bidirectionality and range are paid for in autonomy and infrastructure cost. This pattern repeats across all the technologies of the lecture and is what makes the choice an engineering problem and not a catalogue one.

Slide 18. RPMA (Ingenu) (1/2)



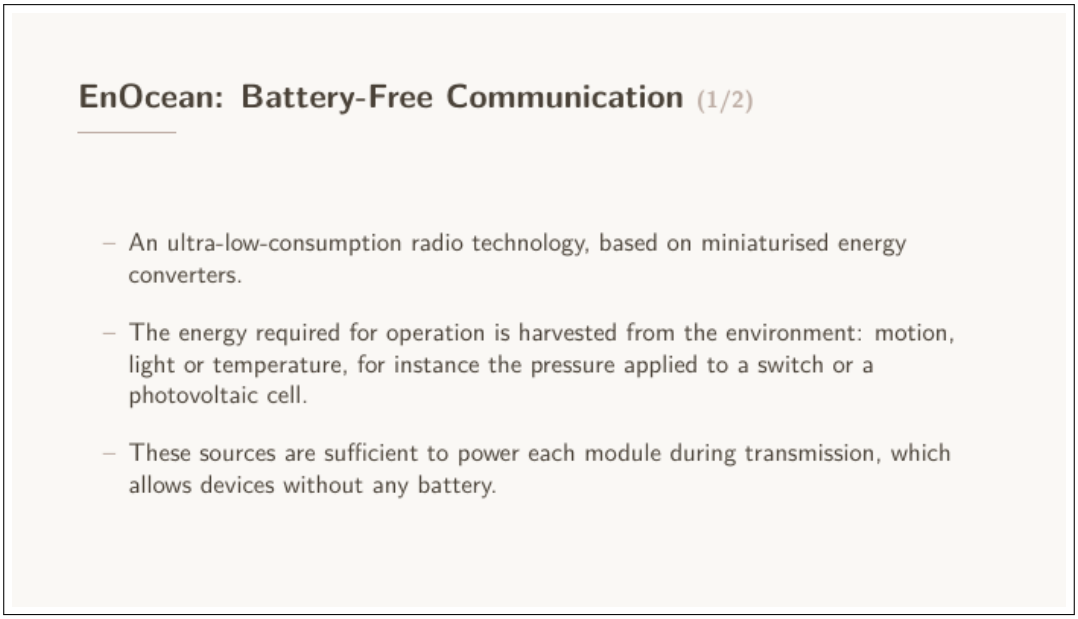
RPMA and the 2.4 GHz band. Random phase multiple access is intended exclusively for machine-to-machine communication and operates at 2.4 GHz, a band available worldwide. That choice simplifies international product certification, but exposes the network to Wi-Fi and Bluetooth interference. Handover support distinguishes RPMA from most LPWANs and makes it applicable to mobile nodes.

Slide 19. RPMA (Ingenu) (2/2)



Physical layer detail. Direct sequence spreading, convolutional coding and Gold codes are the mechanisms that sustain the advertised capacity. Open-loop and closed-loop power control lets near and far nodes coexist without drowning each other out.

Slide 20. EnOcean: Battery-Free Communication (1/2)



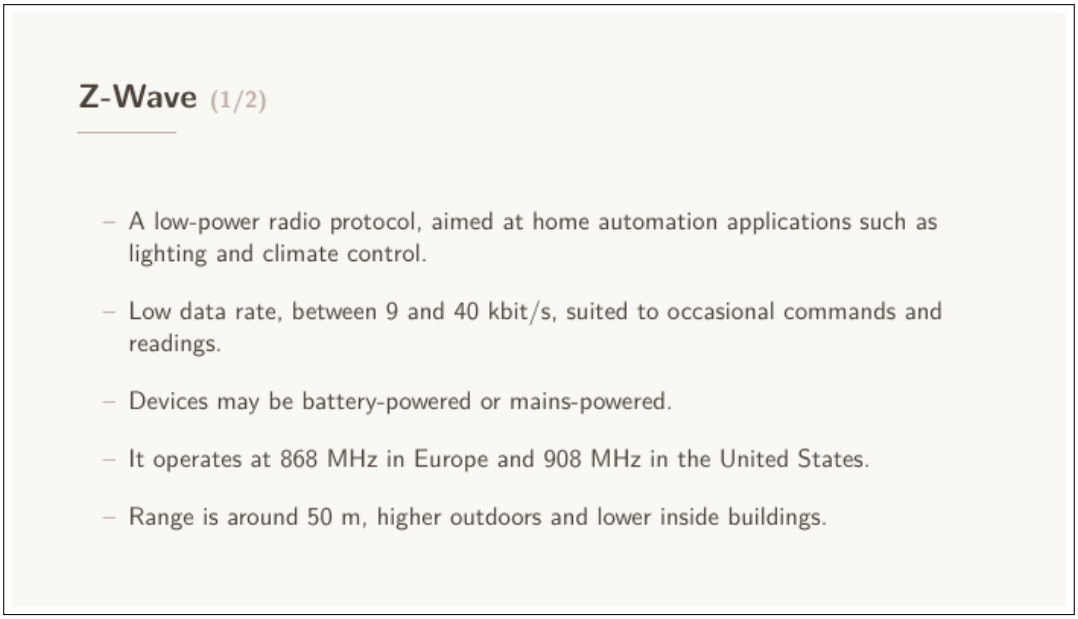
EnOcean: dispensing with the battery. This is the conceptually most distinct technology in the lecture. The operating energy is harvested from the environment, from pressure on a switch, incident light or a temperature gradient. The design implication is considerable: it eliminates the maintenance associated with replacing batteries, which in installations of hundreds of nodes is often the largest operating cost.

Slide 21. EnOcean: Battery-Free Communication (2/2)



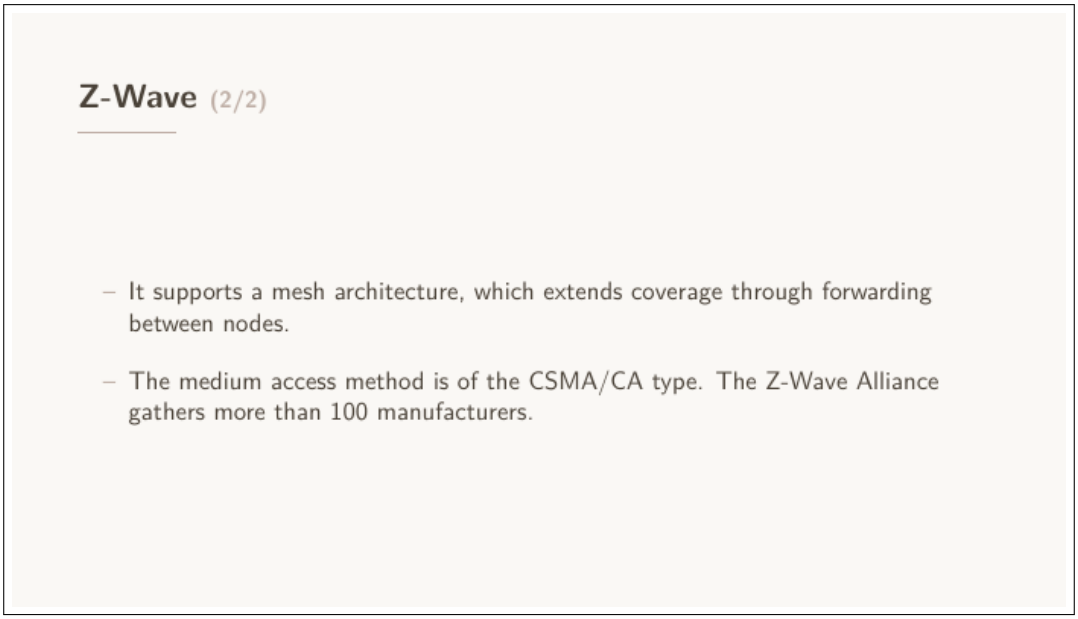
Where it operates and who sustains it. The frequencies, 868 MHz in Europe and 315 MHz in the United States, follow the pattern of the other unlicensed technologies. The size of the EnOcean Alliance indicates industrial maturity, which is a relevant criterion when choosing a technology for an installation with a long service life.

Slide 22. Z-Wave (1/2)



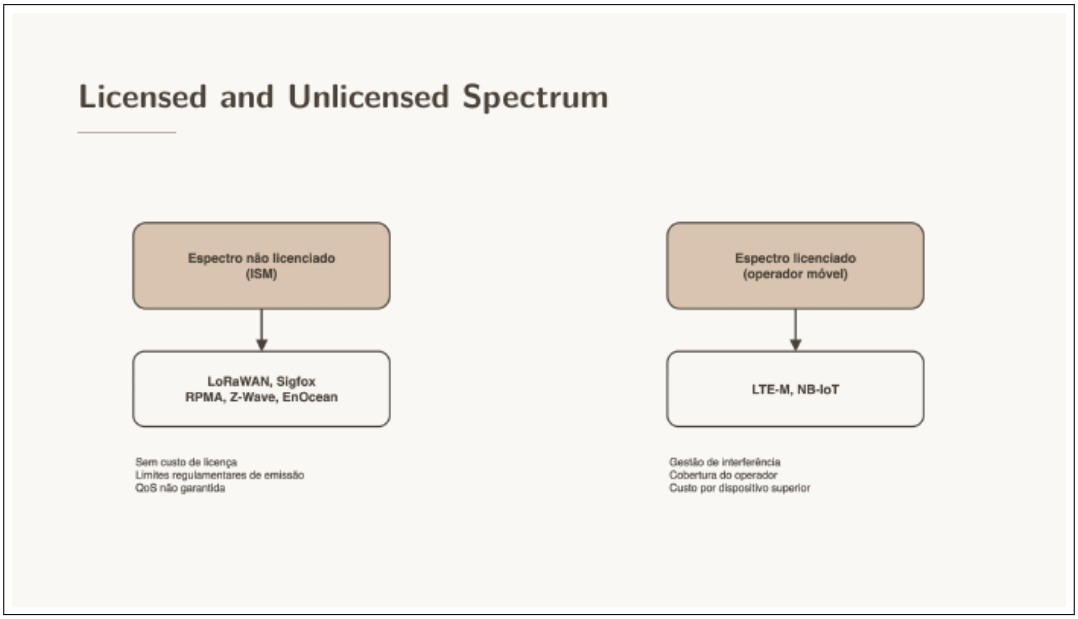
Z-Wave and the home automation domain. Z-Wave has a particular place in this lecture, because its range of about 50 metres puts it outside what is usually called a wide area network. It is presented by comparison, and for its relevance in the home automation market. The data rate between 9 and 40 kbit/s is enough for commands and occasional readings, which is the typical traffic in this domain.

Slide 23. Z-Wave (2/2)



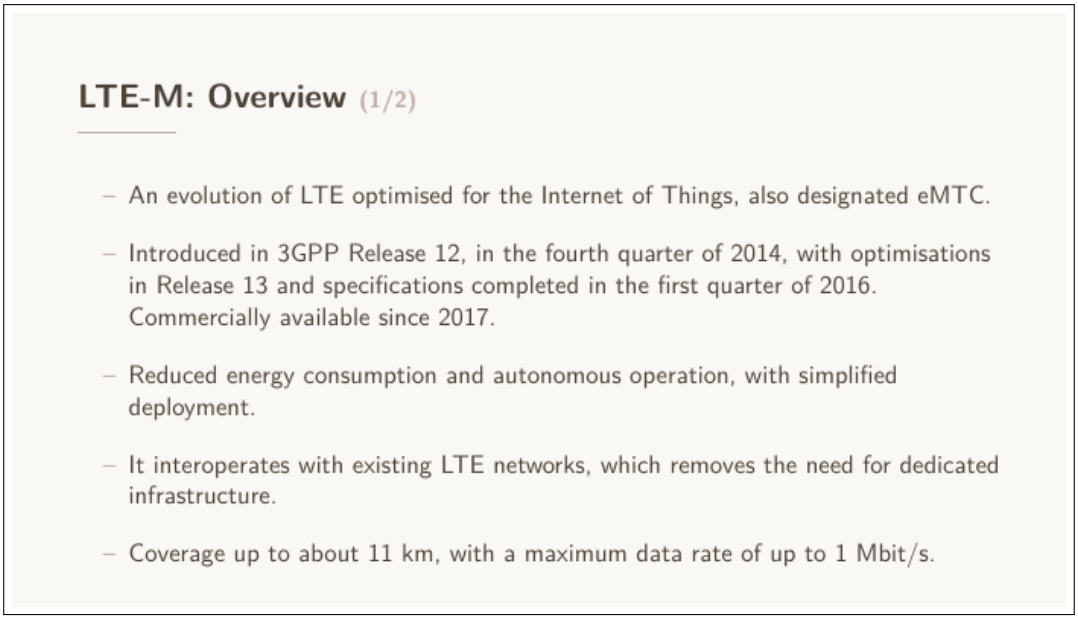
Mesh and medium access. Support for mesh architecture lets coverage be extended by forwarding between nodes, as in Zigbee. The medium access method is CSMA/CA, already studied in Lecture 01. Recognising that the same access mechanism reappears in different technologies is part of the aim of the course.

Slide 24. Licensed and Unlicensed Spectrum



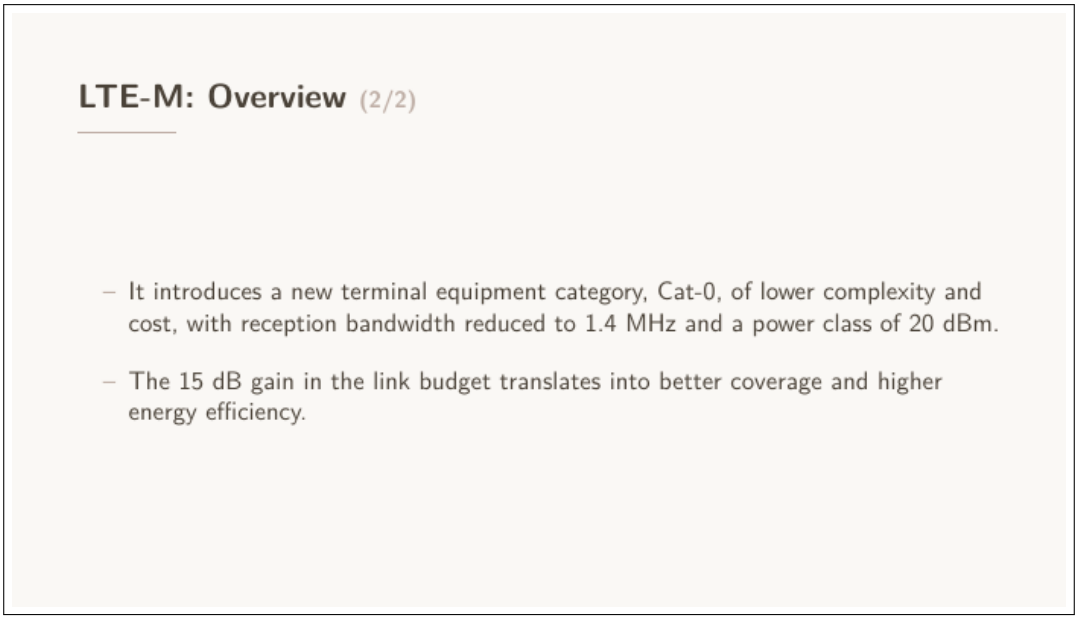
The division that organises the whole lecture. This diagram summarises the most important distinction of the topic. On the left, technologies in unlicensed spectrum, with no licence cost but subject to regulatory emission limits and with no quality-of-service guarantees. On the right, technologies in licensed spectrum, with interference management and coverage by the operator, at a higher cost per device. When a requirement demands guaranteed quality of service, the left half of the diagram is automatically excluded. This is the fastest elimination criterion to apply.

Slide 25. LTE-M: Overview (1/2)



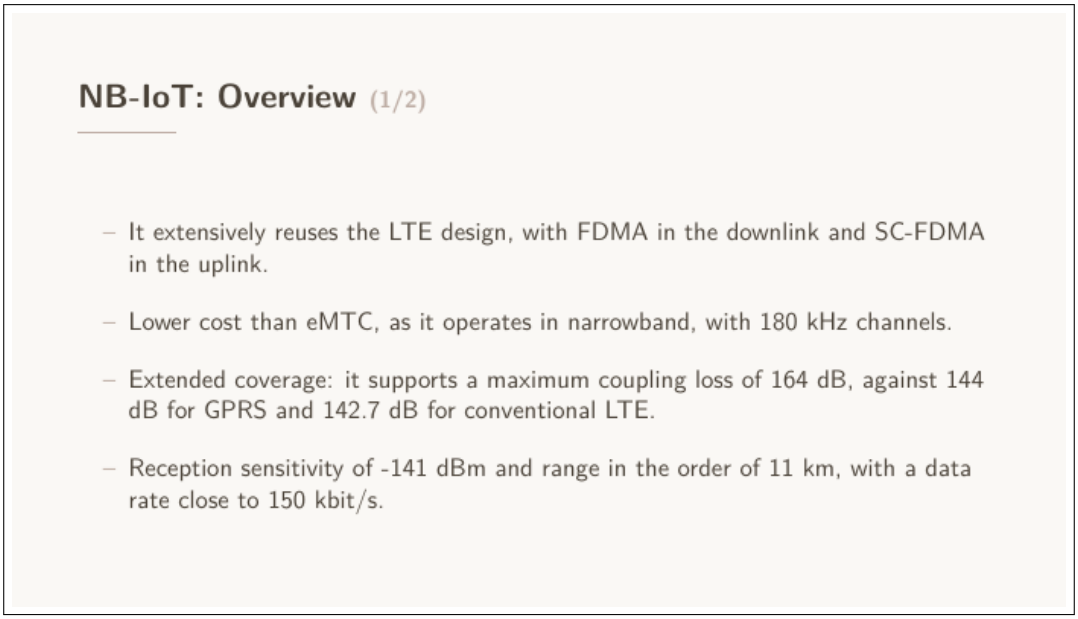
LTE-M as an evolution of LTE. LTE-M, also called eMTC, was introduced in 3GPP Release 12. The decisive property is interoperation with existing LTE networks, which dispenses with dedicated infrastructure. With a data rate of up to 1 Mbit/s, it is the technology of the lecture with the greatest capacity, which makes it the only candidate when requirements include voice or file transfer.

Slide 26. LTE-M: Overview (2/2)



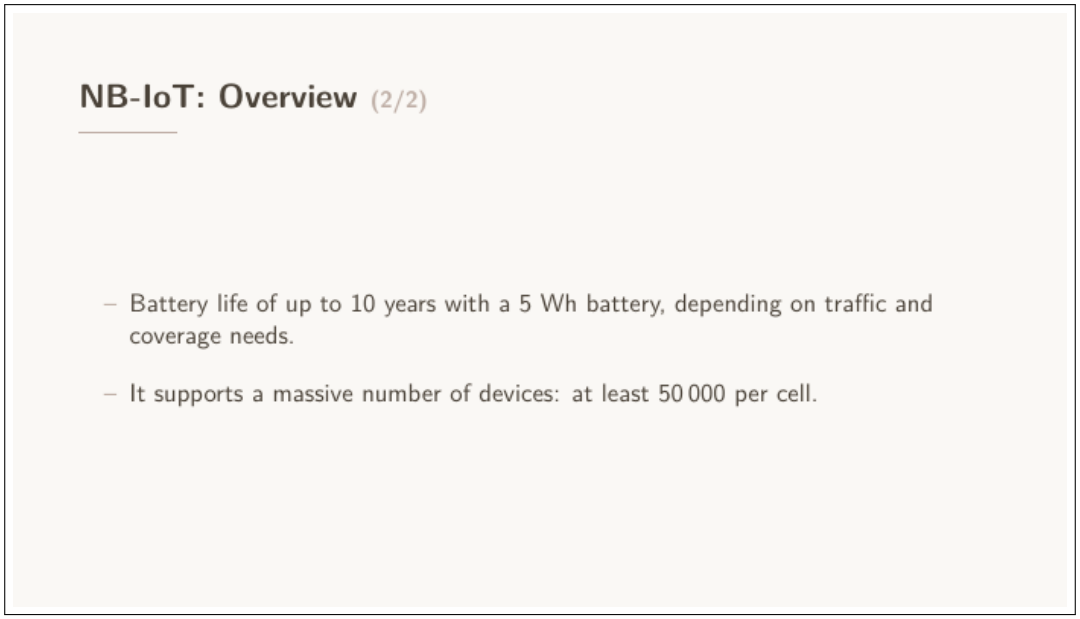
The Cat-0 category and coverage gain. The introduction of a simpler terminal category, with receive bandwidth reduced to 1.4 MHz, is the mechanism by which device cost is reduced without changing the network. The 15 dB gain in the link budget translates into coverage and energy efficiency.

Slide 27. NB-IoT: Overview (1/2)



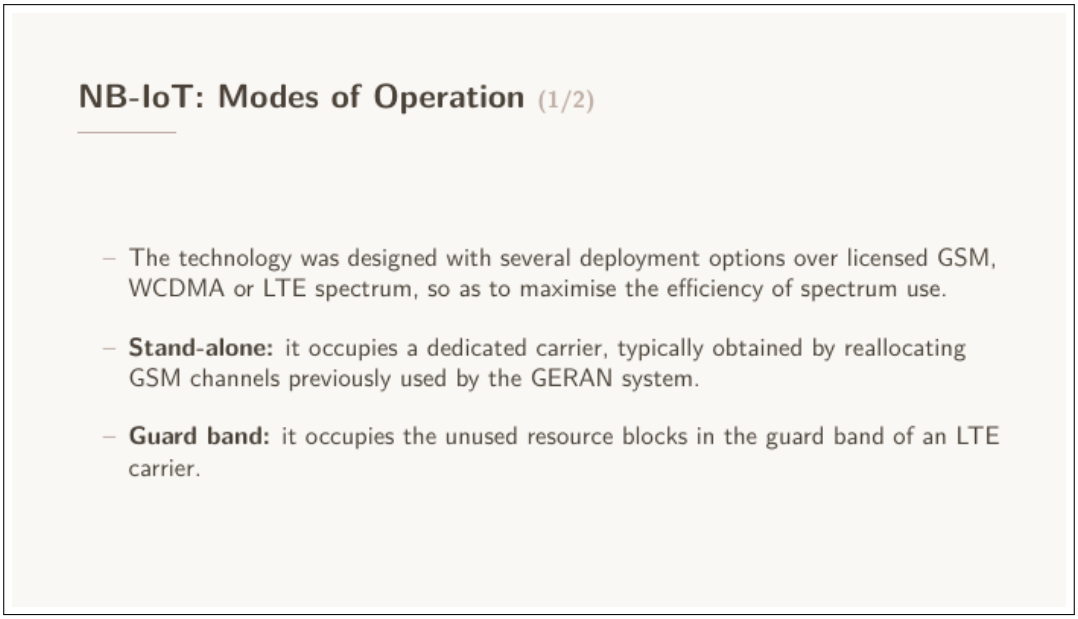
NB-IoT and the choice of narrow band. NB-IoT extensively reuses the LTE design, but operates in 180 kHz channels. The maximum coupling loss supported, 164 dB, compares favourably with the 144 dB of GPRS and the 142.7 dB of conventional LTE. This difference explains why NB-IoT reaches devices in basements and inside ducts, where conventional LTE does not reach.

Slide 28. NB-IoT: Overview (2/2)



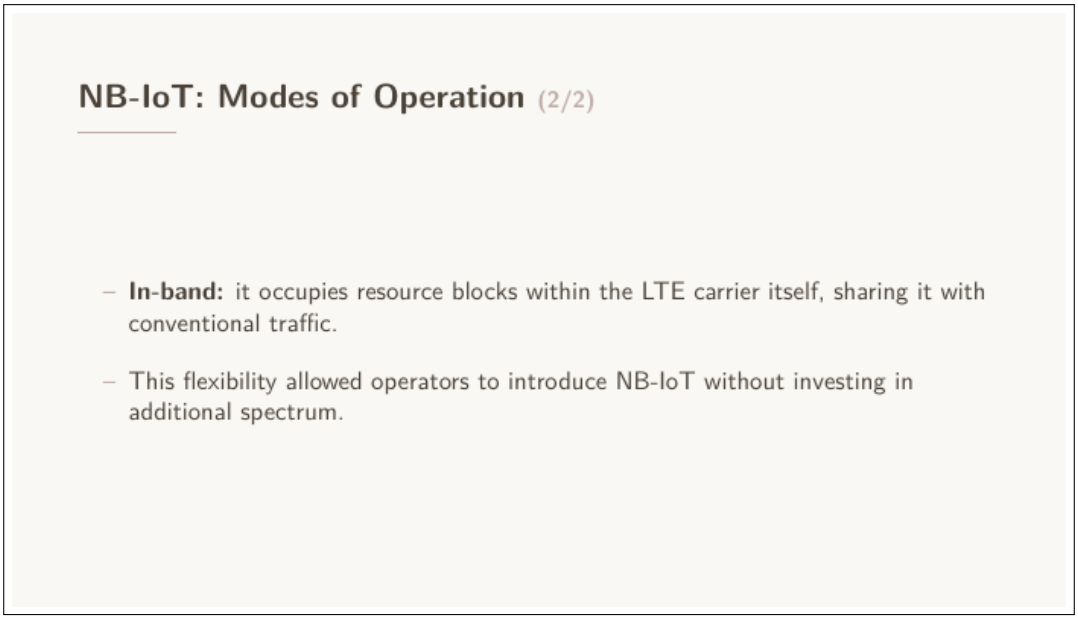
Autonomy and device density. Ten years of autonomy with a 5 Wh battery, and at least 50 thousand devices per cell, are the two numbers that justify the existence of NB-IoT as a technology distinct from LTE-M. The density supported is what makes applications such as city-scale meter reading viable.

Slide 29. NB-IoT: Modes of Operation (1/2)



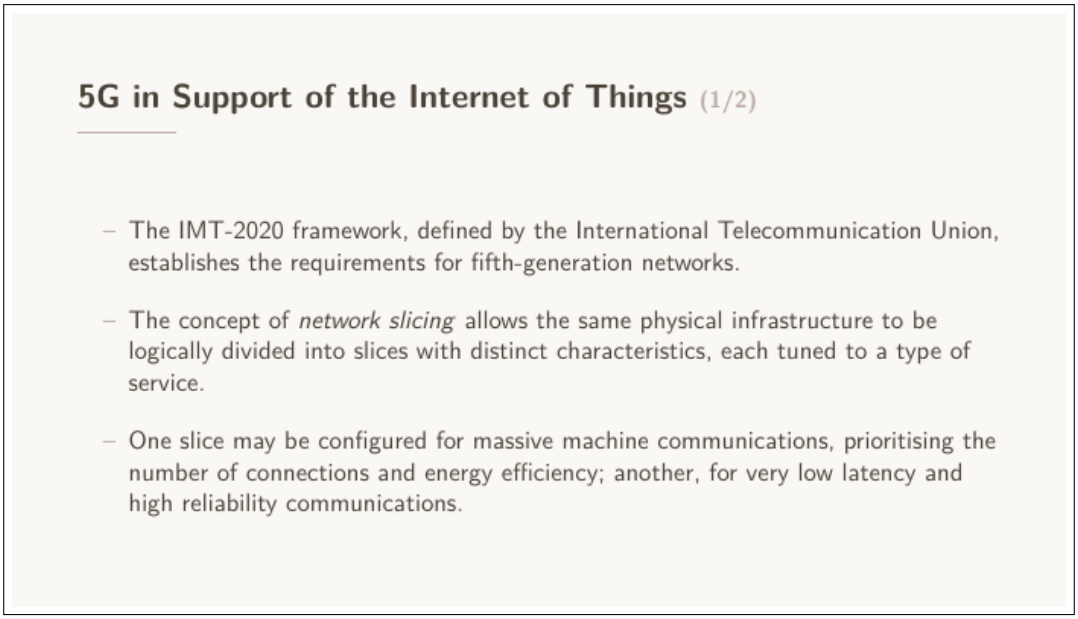
Three ways of finding spectrum. The stand-alone and guard-band modes show the flexibility of deployment. The first re-farms freed GSM carriers. The second uses unused resource blocks in the guard band of an LTE carrier.

Slide 30. NB-IoT: Modes of Operation (2/2)



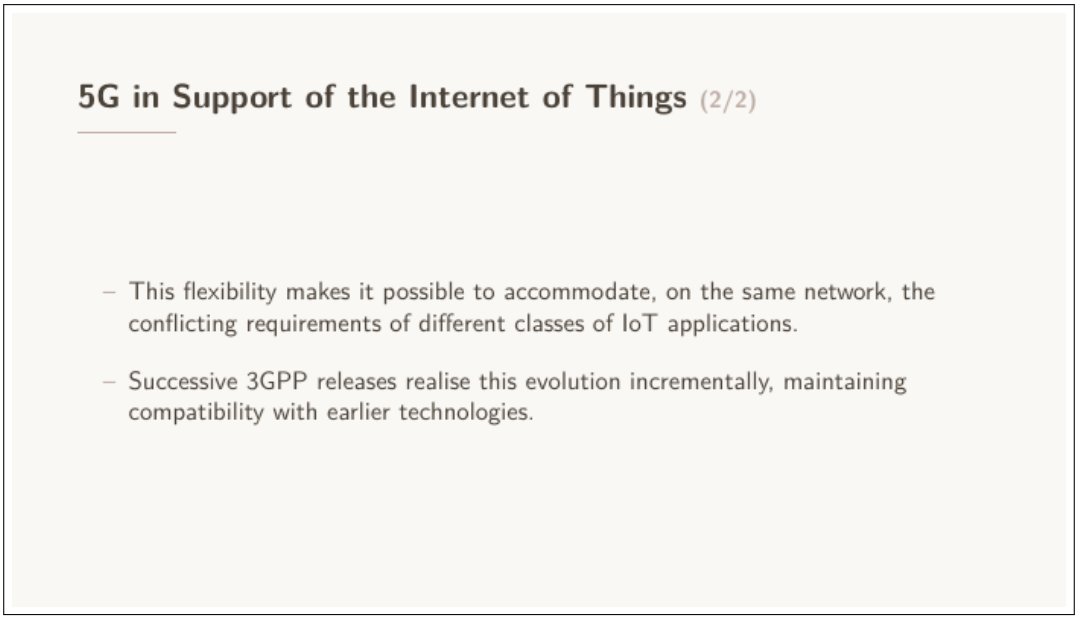
In-band mode and the reason for adoption. In-band mode shares the LTE carrier with conventional traffic. Together, the three modes let operators introduce NB-IoT without acquiring additional spectrum. Understanding this point explains how quickly the technology was deployed, and anticipates the market dominance forecast presented later.

Slide 31. 5G in Support of the Internet of Things (1/2)



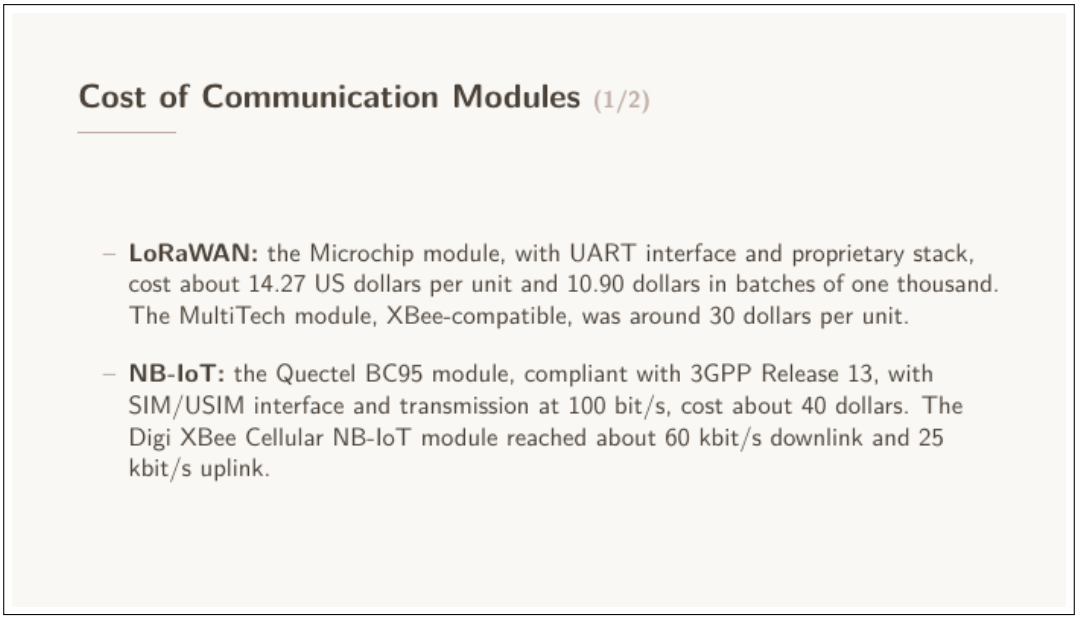
5G and the concept of network slicing. The IMT-2020 framework establishes the requirements of fifth-generation networks. The central concept for this course is slicing, which lets the same physical infrastructure be logically divided into slices with different characteristics. One slice may favour the number of connections and energy efficiency. Another may favour minimum latency and high reliability.

Slide 32. 5G in Support of the Internet of Things (2/2)



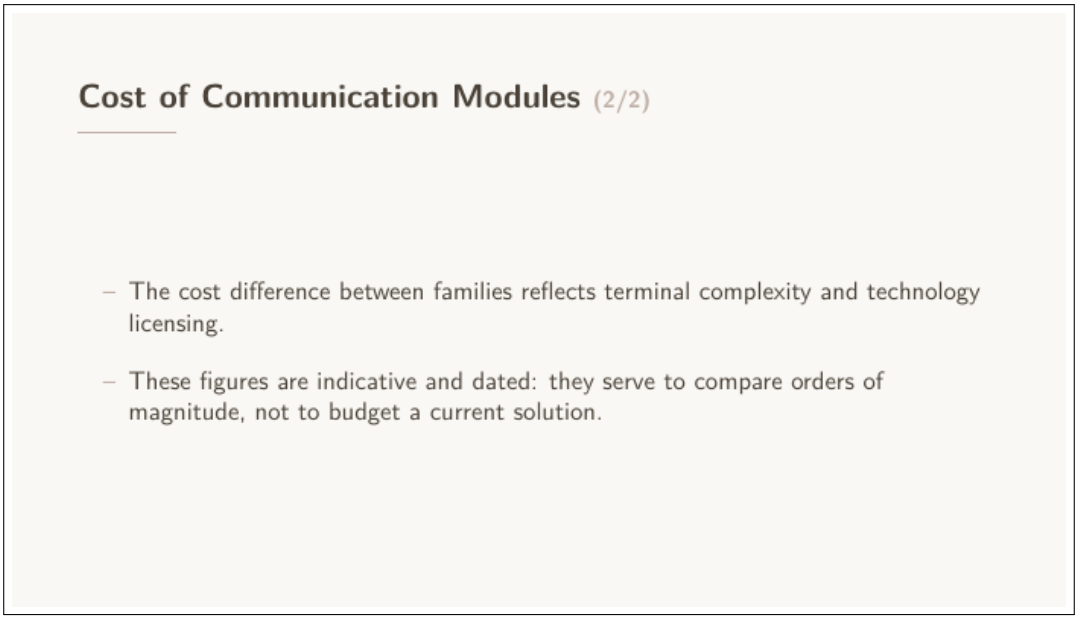
Why slicing solves a real problem. The requirements of the different classes of IoT application are contradictory. Without slicing, a single network would have to be dimensioned for the worst case in each dimension. The successive 3GPP releases realise this evolution incrementally, keeping compatibility with earlier technologies, which is a requirement of a market with long-lived installed equipment.

Slide 33. Cost of Communication Modules (1/2)



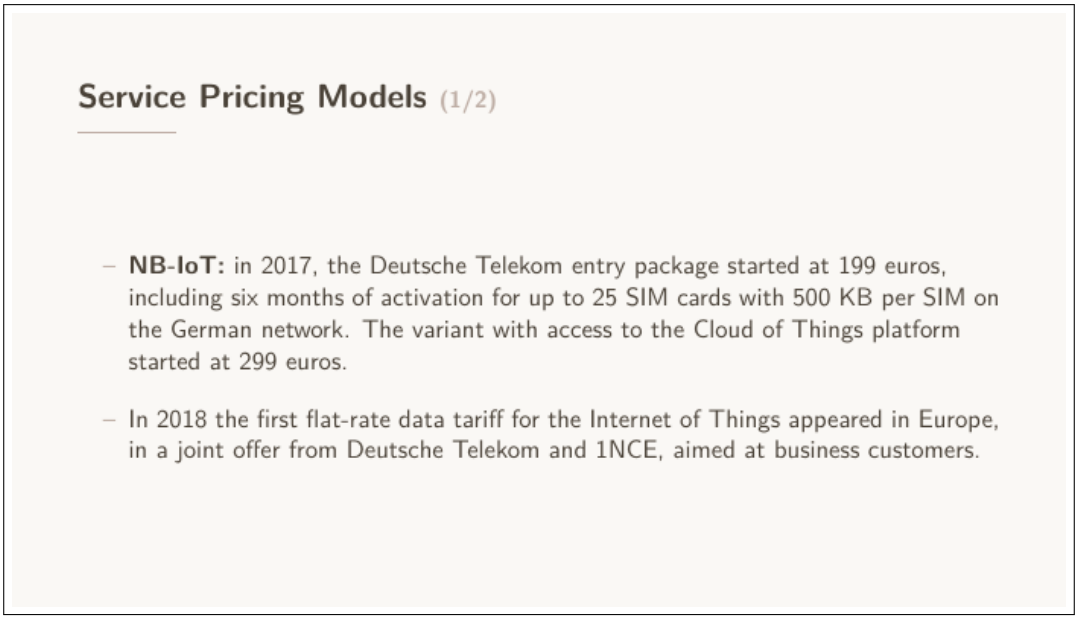
The cost of the communication module. The values shown should be read as orders of magnitude and not as current prices. What matters is the ratio between families: a LoRaWAN module cost about a third of an NB-IoT module. In an installation of thirty thousand devices, that difference determines the viability of the project.

Slide 34. Cost of Communication Modules (2/2)



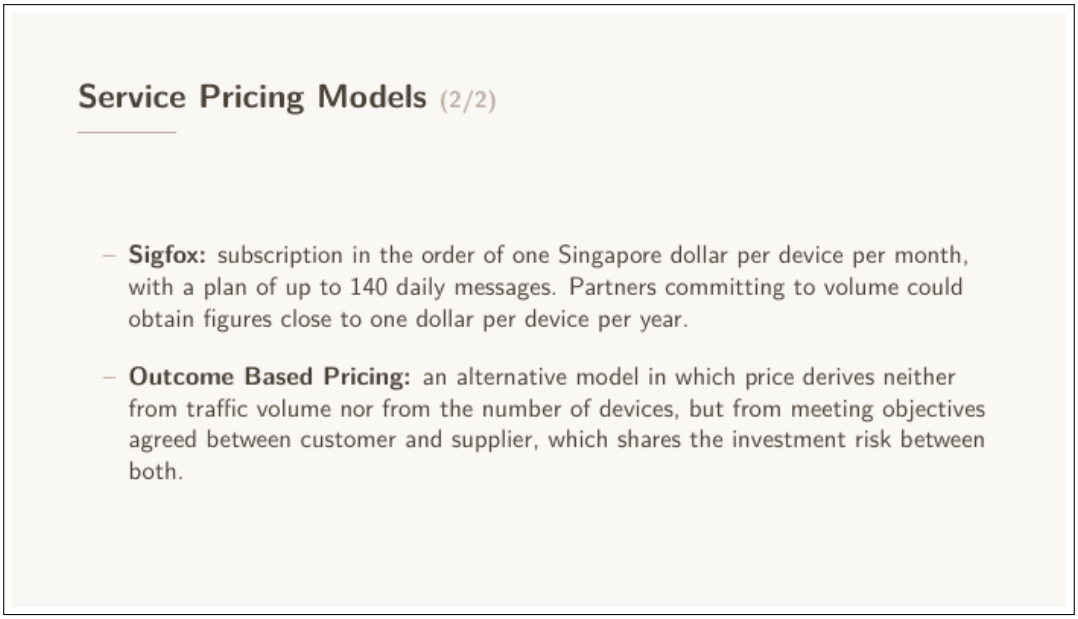
Reading dated numbers carefully. The cost difference reflects the complexity of the terminal and the licensing of the technology. The slide explicitly states that the values are indicative and dated. This warning is part of the material. Knowing how to qualify the temporal validity of a technical figure is an engineering skill.

Slide 35. Service Pricing Models (1/2)



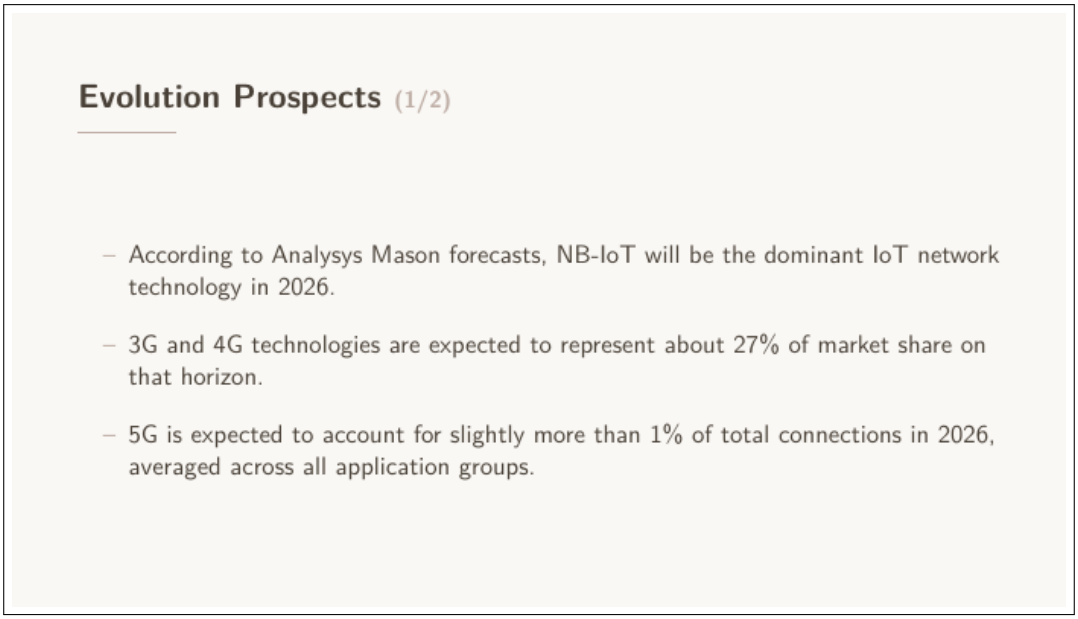
Subscription models. Entry packages and flat rates illustrate how cost shifts from equipment to service. In operator technologies, the total cost of ownership is dominated by the subscription over the years, not by the initial price of the module.

Slide 36. Service Pricing Models (2/2)



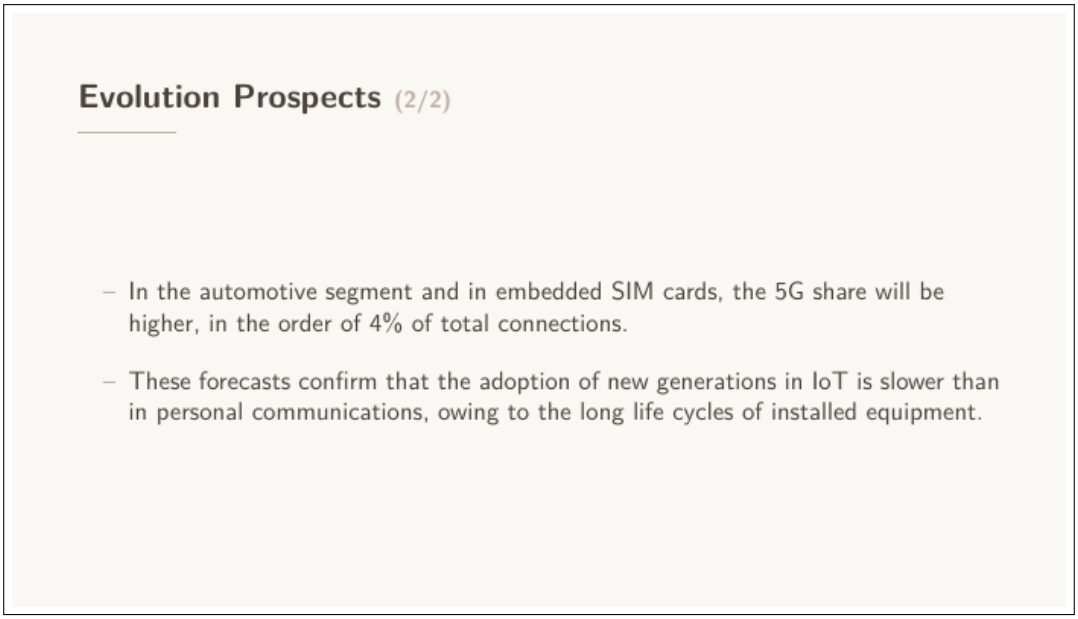
Outcome-based pricing. The Outcome Based Pricing model breaks with the logic of volume. The price does not follow from traffic or the number of devices, but from meeting agreed objectives. The consequence is the sharing of investment risk between customer and supplier, which changes the nature of the commercial relationship and sometimes makes viable projects that would otherwise have no funding.

Slide 37. Evolution Prospects (1/2)



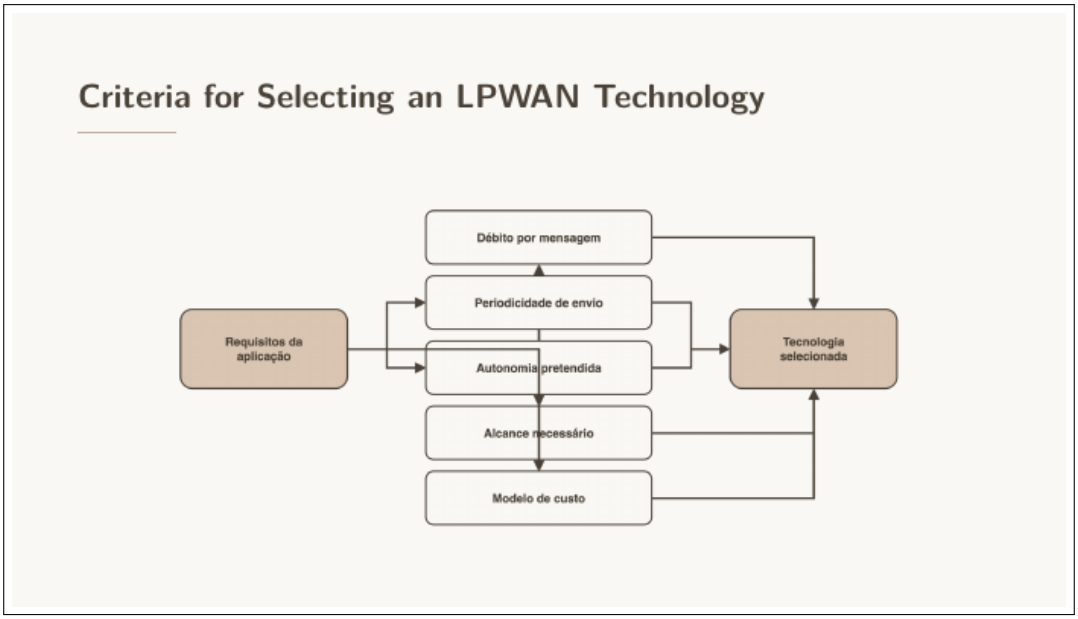
Where the market is heading. The forecast of NB-IoT dominance in 2026, with 3G and 4G representing about 27 per cent, should be read as a trend and not as a certainty. The most interesting figure is the small share attributed to 5G, just over 1 per cent of total connections.

Slide 38. Evolution Prospects (2/2)



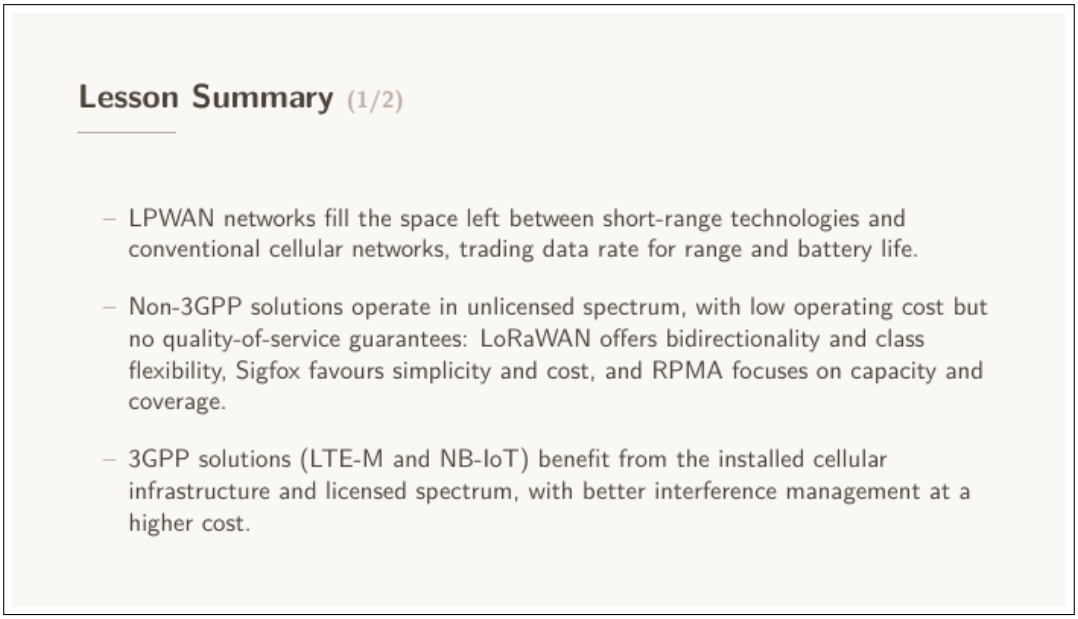
Why adoption in IoT is slow. The explanation lies in the life cycles of installed equipment. A meter installed today will stay in service for a decade or more, and will not be replaced just because a new generation has appeared. This inertia is a structural feature of the sector and should be considered in any design decision.

Slide 39. Criteria for Selecting an LPWAN Technology



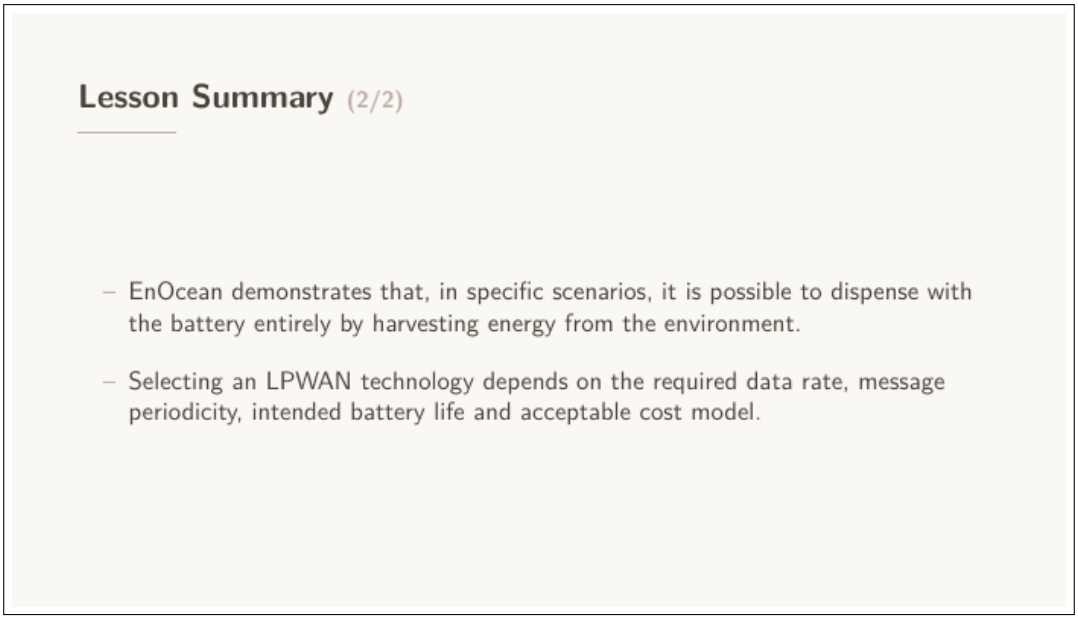
The criteria that structure the decision. The diagram gathers the five criteria that allow moving from application requirements to a concrete technology: data per message, sending periodicity, desired autonomy, required range and cost model. This is the operational content of the lecture and is directly exercised in the lab session, where a tool that applies these criteria to real scenarios is built.

Slide 40. Lesson Summary (1/2)



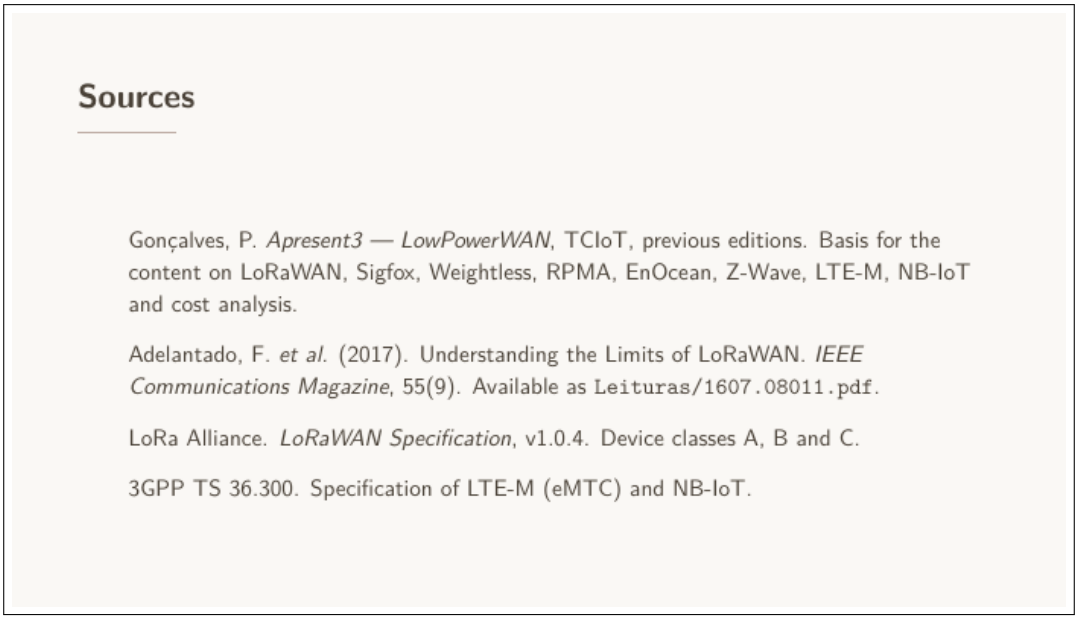
Summary, first part. LPWANs fill the space between short range and conventional cellular networks, trading data rate for range and autonomy. Non-3GPP solutions have low operating cost but offer no quality-of-service guarantees. Within that family, each technology optimises a different dimension: LoRaWAN flexibility, Sigfox simplicity and cost, RPMA capacity.

Slide 41. Lesson Summary (2/2)



Summary, second part. EnOcean shows that, in specific scenarios, it is possible to dispense with the battery altogether. The final conclusion of the lecture is the one worth fixing: selection depends on the data rate required, the periodicity of messages, the desired autonomy and the acceptable cost model. There is no best technology, there is the one that satisfies a concrete set of constraints.

Slide 42. Sources



Origin of the material. The sources identify the provenance of the content. The basis is the course material from previous editions. The paper by Adelantado and co-authors, available in the readings folder, goes deeper into the limits of LoRaWAN and is recommended reading for anyone who wants to go beyond the lecture.

Final Check

Before considering the material studied, check that you can answer the following without looking at the slides.

1. Distinguish LoRa from LoRaWAN, stating which layer each term corresponds to.
2. State, for the three LoRaWAN device classes, the compromise between consumption and downlink latency.
3. Explain why Sigfox does not serve an application that requires delivery confirmation.
4. List the practical consequences of operating in licensed spectrum, as opposed to unlicensed spectrum.
5. Justify the coexistence of LTE-M and NB-IoT, even though both are 3GPP solutions.
6. Explain why the adoption of new generations is slower in IoT than in personal communications.
7. Apply the five selection criteria to a given scenario, and justify the exclusion of each discarded technology.

References

- Gonçalves, P. *Apresent3 — LowPowerWAN*, TCIoT, previous editions.
- Adelantado, F., Vilajosana, X., Tuset-Peiro, P., Martinez, B., Melià-Seguí, J. and Watteyne, T. (2017). Understanding the Limits of LoRaWAN. *IEEE Communications Magazine*, 55(9), 34–40.
- LoRa Alliance. *LoRaWAN Specification*, v1.0.4.
- 3GPP TS 36.300. Specification of LTE-M (eMTC) and NB-IoT.